

Distributed [*Computing*] Systems

Interconnection Networks, Communications and the OSI Model - Part II

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The OSI Reference Model



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Source

- ▶ Source: http://www.dcs.napier.ac.uk/~bill/cisco_presentation/osi31.ppt



Overview

- ▶ The need for standards
- ▶ Osi - organisation for standardisation
- ▶ The osi reference model
- ▶ A layered network model
- ▶ The seven osi reference model layers
- ▶ Summary



The Need For Standards

- ▶ Over the past couple of decades many of the networks that were built used different hardware and software implementations, as a result they were incompatible and it became difficult for networks using different specifications to communicate with each other.
- ▶ To address the problem of networks being incompatible and unable to communicate with each other, the International Organisation for Standardisation (ISO) researched various network schemes.
- ▶ The ISO recognized there was a need to create a **NETWORK MODEL** that would help vendors create interoperable network implementations.



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ISO - Organisation For Standardisation

- ▶ The International Organisation for Standardisation (ISO) is an International standards organisation responsible for a wide range of standards, including many that are relevant to networking.
- ▶ In 1984 in order to aid network interconnection without necessarily requiring complete redesign, the Open Systems Interconnection (OSI) reference model was approved as an international standard for communications architecture.



International Organization for Standardization



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The OSI Reference Model

- ▶ The model was developed by the International Organisation for Standardisation (ISO) in 1984. It is now considered the primary Architectural model for inter-computer communications.
- ▶ The Open Systems Interconnection (OSI) reference model is a descriptive network scheme. It ensures greater compatibility and interoperability between various types of network technologies.
- ▶ The OSI model describes how information or data makes its way from application programmes (such as spreadsheets) through a network medium (such as wire) to another application programme located on another network.
- ▶ The OSI reference model divides the problem of moving information between computers over a network medium into SEVEN smaller and more manageable problems.
- ▶ This separation into smaller more manageable functions is known as layering.

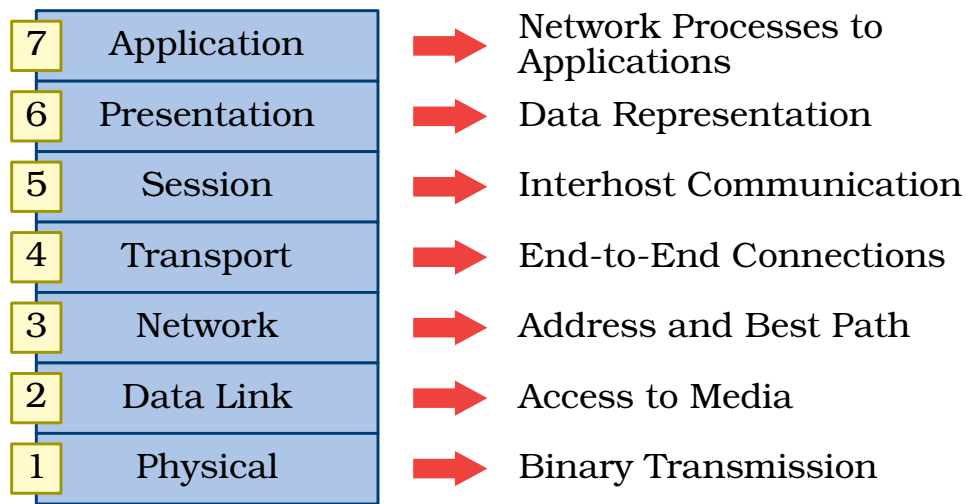


A Layered Network Model

- ▶ The OSI Reference Model is composed of seven layers, each specifying particular network functions.
- ▶ The process of breaking up the functions or tasks of networking into layers reduces complexity.
- ▶ Each layer provides a service to the layer above it in the protocol specification.
- ▶ Each layer communicates with the same layer's software or hardware on other computers.
- ▶ The lower 4 layers (transport, network, data link and physical—Layers 4, 3, 2, and 1) are concerned with the flow of data from end to end through the network.
- ▶ The upper four layers of the OSI model (application, presentation and session—Layers 7, 6 and 5) are orientated more toward services to the applications.
- ▶ Data is Encapsulated with the necessary protocol information as it moves down the layers before network transit.



The Seven OSI reference model layers



Layer 7: Application

- ▶ The application layer is the OSI layer that is closest to the user.
- ▶ It provides network services to the user's applications.
- ▶ It differs from the other layers in that it does not provide services to any other OSI layer, but rather, only to applications outside the OSI model.
- ▶ Examples of such applications are spreadsheet programs, word processing programs, and bank terminal programs.
- ▶ The application layer establishes the availability of intended communication partners, synchronizes and establishes agreement on procedures for error recovery and control of data integrity.



Layer 6: Presentation

- ▶ The presentation layer ensures that the information that the application layer of one system sends out is readable by the application layer of another system.
- ▶ If necessary, the presentation layer translates between multiple data formats by using a common format.
- ▶ Provides encryption and compression of data.
- ▶ Examples :- JPEG, MPEG, ASCII, EBCDIC, HTML.



Layer 5: Session

- ▶ The session layer defines how to start, control and end conversations (called sessions) between applications.
- ▶ This includes the control and management of multiple bi-directional messages using dialogue control.
- ▶ It also synchronizes dialogue between two hosts' presentation layers and manages their data exchange.
- ▶ The session layer offers provisions for efficient data transfer.
- ▶ Examples :- SQL, ASP(AppleTalk Session Protocol).



Layer 4: Transport

- ▶ The transport layer regulates information flow to ensure end-to-end connectivity between host applications reliably and accurately.
- ▶ The transport layer segments data from the sending host's system and reassembles the data into a data stream on the receiving host's system.
- ▶ The boundary between the transport layer and the session layer can be thought of as the boundary between application protocols and data-flow protocols. Whereas the application, presentation, and session layers are concerned with application issues, the lower four layers are concerned with data transport issues.
- ▶ Layer 4 protocols include TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).



Layer 3: Network

- ▶ Defines end-to-end delivery of packets.
- ▶ Defines logical addressing so that any endpoint can be identified.
- ▶ Defines how routing works and how routes are learned so that the packets can be delivered.
- ▶ The network layer also defines how to fragment a packet into smaller packets to accommodate different media.
- ▶ Routers operate at Layer 3.
- ▶ Examples :- IP, IPX, AppleTalk.



Layer 2: Data Link

- ▶ The data link layer provides access to the networking media and physical transmission across the media and this enables the data to locate its intended destination on a network.
- ▶ The data link layer provides reliable transit of data across a physical link by using the Media Access Control (MAC) addresses.
- ▶ The data link layer uses the MAC address to define a hardware or data link address in order for multiple stations to share the same medium and still uniquely identify each other.
- ▶ Concerned with network topology, network access, error notification, ordered delivery of frames, and flow control.
- ▶ Examples :- Ethernet, Frame Relay, FDDI.



Layer 1: Physical

- ▶ The physical layer deals with the physical characteristics of the transmission medium.
- ▶ It defines the electrical, mechanical, procedural, and functional specifications for activating, maintaining, and deactivating the physical link between end systems.
- ▶ Such characteristics as voltage levels, timing of voltage changes, physical data rates, maximum transmission distances, physical connectors, and other similar attributes are defined by physical layer specifications.
- ▶ Examples :- EIA/TIA-232, RJ45, NRZ.

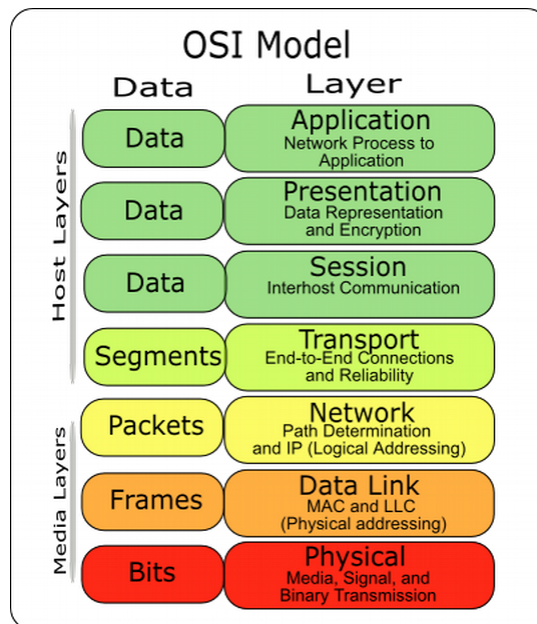


Summary

- ▶ There was no standard for networks in the early days and as a result it was difficult for networks to communicate with each other.
- ▶ The International Organisation for Standardisation (ISO) recognised this. and researched various network schemes, and in 1984 introduced the Open Systems Interconnection (OSI) reference model.
- ▶ The OSI reference model has standards which ensure vendors greater compatibility and interoperability between various types of network technologies.
- ▶ The OSI reference model organizes network functions into seven numbered layers.
- ▶ Each layer provides a service to the layer above it in the protocol specification and communicates with the same layer's software or hardware on other computers.
- ▶ Layers 1-4 are concerned with the flow of data from end to end through the network and Layers 5-7 are concerned with services to the applications.



The OSI Model



Layer 1: Physical Layer

- ▶ Handles bit level communication across the network channel.
- ▶ Defines specifications for activating, maintaining and deactivating the physical link between connecting network systems.
- ▶ Standards: RS232, T1, DSL, 10BASE-T
- ▶ Defines physical means of sending data over network devices
- ▶ Interfaces between network medium and devices
- ▶ Defines optical, electrical and mechanical characteristics



Layer 2: Data Link Layer

- ▶ Controls how a computer gains access to the data and permission to transmit it (MAC sub layer).
- ▶ Synchronizes frames, performs flow control and error checking (LLC sub layer).
- ▶ This layer has the important task of creating and managing what frames are sent out on the network.
- ▶ Protocols: Ethernet, PPP, MPLS, ATM, Frame Relay
- ▶ Defines procedures for operating the communication links
- ▶ Frames packets
- ▶ Detects and corrects packets transmit errors



Layer 3: Network Layer

- ▶ Provides means to establish, maintain and terminate network connections.
- ▶ Main functions performed:
 - ▶ Addressing and delivery of packets
 - ▶ Routing – determining the path that the data should take.
- ▶ Provides an abstraction of the network to the transport layer, either as a Virtual Circuit or a Datagram.
- ▶ IP lives in this layer!
- ▶ Other Protocols: ICMP, IPX, IPSec
- ▶ This layer handles the routing of the data
 - ▶ Determines how data are transferred between network devices.
 - ▶ Routes packets according to unique network device addresses (sending it in the right direction to the right destination on outgoing transmissions and receiving incoming transmissions at the packet level).
 - ▶ Provides flow and congestion control to prevent network resource depletion.
 - ▶ The network layer does routing and forwarding.



Layer 4: Transport Layer

- ▶ Provides transparent transfer of data between end users
- ▶ Provides reliable and cost-effective transfer of data
 - ▶ Connection-oriented (TCP)
 - ▶ Connectionless (UDP)
- ▶ This layer manages the end-to-end and error-checking.
 - ▶ Manages end-to-end message delivery in network
 - ▶ Provides reliable and sequential packet delivery through error recovery and flow control mechanisms (for example, determining whether all packets have arrived)
 - ▶ Provides connectionless oriented packet delivery
 - ▶ Ensures complete data transfer



Layer 5: Session Layer (Enhanced Version Of The Transport Layer)

- ▶ Provides dialog control between end-user application processes (duplex or half-duplex operations)
- ▶ Establishes checkpointing, adjournment, termination, and restart procedures
- ▶ Sets up and tears down TCP/IP sessions
- ▶ This layer sets up, coordinates, and terminates conversations, exchanges, and dialogs between the applications at each end.
 - ▶ Manages user sessions and dialogues
 - ▶ Controls establishment and termination of logic links between users
 - ▶ Reports upper layer errors



Layer 6: Presentation Layer

- ▶ Relieves the Application layer of concern regarding syntactical differences in data representation within the end-user systems.
 - ▶ Multipurpose Internet Mail Extensions (MIME)
 - ▶ Data compression
 - ▶ Encryption
 - ▶ Character conversions
- ▶ This is a layer, usually part of an operating system, that converts incoming and outgoing data from one presentation format to another
 - ▶ Masks the differences of data formats between dissimilar systems
 - ▶ Specifies architecture-independent data transfer format
 - ▶ Encodes and decodes data; Encrypts and decrypts data; Compresses and decompresses data



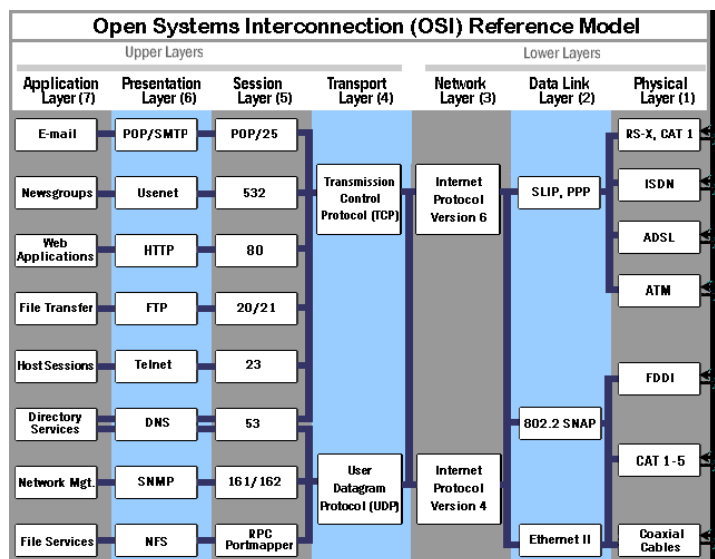
Layer 7: Application Layer

- ▶ Facilitates communication between software applications
- ▶ Mediates negotiations between applications:
 - ▶ Formatting
 - ▶ Security
 - ▶ Synchronization
- ▶ HTTP – defines protocol for remote management and transfer handling of Web pages (web browsers and web servers)
- ▶ FTP – defines protocol for transferring files between a client and server machine
- ▶ This is the layer at which communication partners are identified, quality of service is identified, user authentication and privacy are considered, and any constraints on data syntax are identified.
 - ▶ Defines interface to user processes for communication and data transfer in network
 - ▶ Provides standardized services such as virtual terminal, file and job transfer and operations



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